

Task A: Statistical Analysis of Hotel Revenue and Implications for Sri Lanka’s Tourism Sector

1. Introduction

The tourism and hospitality sector is a key pillar of Sri Lanka’s economy, contributing significantly to GDP, employment, and foreign exchange. Understanding the drivers of hotel revenue is critical for evidence-based policy and strategic decision-making.

This study analyzes a dataset of 500 hotels to achieve the following objectives:

- 1. Identify and quantify the relationships between operational, marketing, and service quality factors and hotel revenue.
- 2. Develop predictive models to inform strategic decision-making.
- 3. Provide actionable insights contextualized to Sri Lanka’s diverse hospitality landscape, including cultural tourism, beach resorts, and eco-tourism.

The analysis follows rigorous statistical protocols, including normality testing, correlation analysis, linear regression modeling, and diagnostic validation.

2. Data description and methodology

2.1. Dataset Description

The analysis uses the Hotels_2025 dataset with 500 complete observations. While the data reflects global hospitality trends, findings are interpreted for the Sri Lankan market.

2.2. Variable Definitions

Variable	Type	Description	Unit / Scale
Revenue	Continuous	Total monthly hotel revenue (dependent variable)	USD (\$)
Rooms Available	Continuous	Rooms count in the hotel	Count (50–299)

Occupancy Rate	Continuous	Proportion of rooms occupied per month	Ratio (0.40–1.00)
ADR	Continuous	Average Daily Rate per occupied room	USD (\$)
Marketing Spend	Continuous	Monthly marketing expenditure	USD (\$)
Staff Count	Continuous	Number of hotel employees	Count
Guest Satisfaction Score	Continuous	Average monthly guest rating	Scale (2.8–10.0)
Loyalty Members	Continuous	Number of loyalty program members	Count
Hotel Quality Rank	Categorical	Quality classification based on ADR and occupancy	Low / Medium / High

Table 1 Variable definitions

2.3. Descriptive Statistics

The dataset comprises 500 hotels with no missing values. Numeric variables are summarized below

Variable	Mean	Standard Deviation	Min	Q1	Median	Q3	Max
Revenue (USD)	461,798	219,650	86,484	285,580	435,403	617,770	1,310,206
Rooms Available	178	71.77	50	114	179	239	299
Occupancy Rate (%)	70.38	10.76	40	63	71	77	100
ADR (USD)	121.46	20.39	65.75	109.16	121.77	135.20	182.84
Marketing Spend (USD)	19,783	5,073	3,058	16,380	19,666	23,392	33,146
Staff Count	35.41	14.40	5	23	36	48	64
Guest Satisfaction Score	6.80	1.17	2.8	6.1	6.8	7.6	10.0
Loyalty Members	2,526	1,161	180	1,600	2,480	3,409	6,117

Table 2 Descriptive statistics

The distribution of hotel quality ranks shows that 63.4% of hotels are Medium, 26.6% are Low, and 10.0% are High.

2.4. Statistical Methodology

The analysis included:

1. Normality tests (Anderson-Darling, Shapiro-Wilk, Kolmogorov-Smirnov)
2. Correlation analysis (Pearson, Spearman)
3. Simple and multiple linear regression, including interaction terms
4. Model selection based on AIC, BIC, and Adjusted R^2

All analyses were conducted in **R** with significance thresholds: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

3. Hypothesis Based Normality Assessment

Normality testing is a prerequisite for many parametric statistical analyses, which assume that the data follow a normal distribution. To assess normality for all continuous variables, the following tests were conducted:

1. **Shapiro-Wilk Test**
2. **Anderson-Darling Test**
3. **Lilliefors (Kolmogorov-Smirnov) Test**

Graphical assessments using **QQ plots** and **histograms with overlaid normal curves** were also used to visually confirm distribution characteristics.

3.1. Hypotheses

For each normality test, the hypotheses are defined as:

- **Null Hypothesis (H_0):** The data is normally distributed
- **Alternative Hypothesis (H_1):** The data is not normally distributed

A p-value < 0.05 indicates rejection of H_0 , suggesting that the variable deviates from normality.

3.2. Normality tests results

Variable	Shapiro-Wilk p-value	Anderson-Darling p-value	Lilliefors p-value	Skewness	Kurtosis	Normal?
Rooms Available	3.45e-11	1.35e-13	1.53e-07	-0.076	1.822	No
Occupancy Rate	0.270	0.118	0.039	0.044	2.900	Yes
ADR	0.347	0.320	0.342	-0.071	3.093	Yes
Marketing Spend	0.548	0.441	0.297	0.011	2.993	Yes
Staff Count	1.11e-08	2.81e-10	2.29e-06	-0.051	1.930	No
Revenue	3.71e-09	1.19e-08	4.08e-05	0.618	3.142	No
Guest Satisfaction	0.103	0.042	0.059	-0.168	3.026	Yes
Loyalty Members	8.76e-06	1.08e-05	0.00021	0.305	2.485	No

Table 3 Normality Test Results

3.3. Graphical assessment

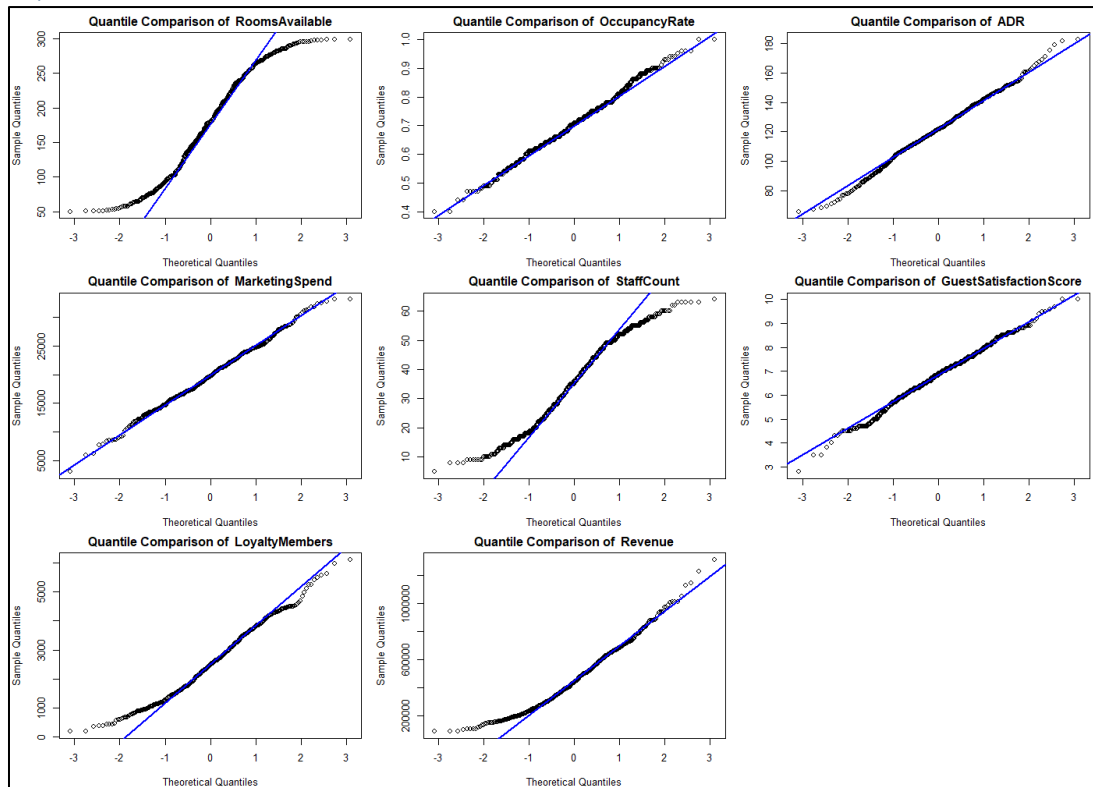


Figure 1 QQ Plots for All Numerical Variables

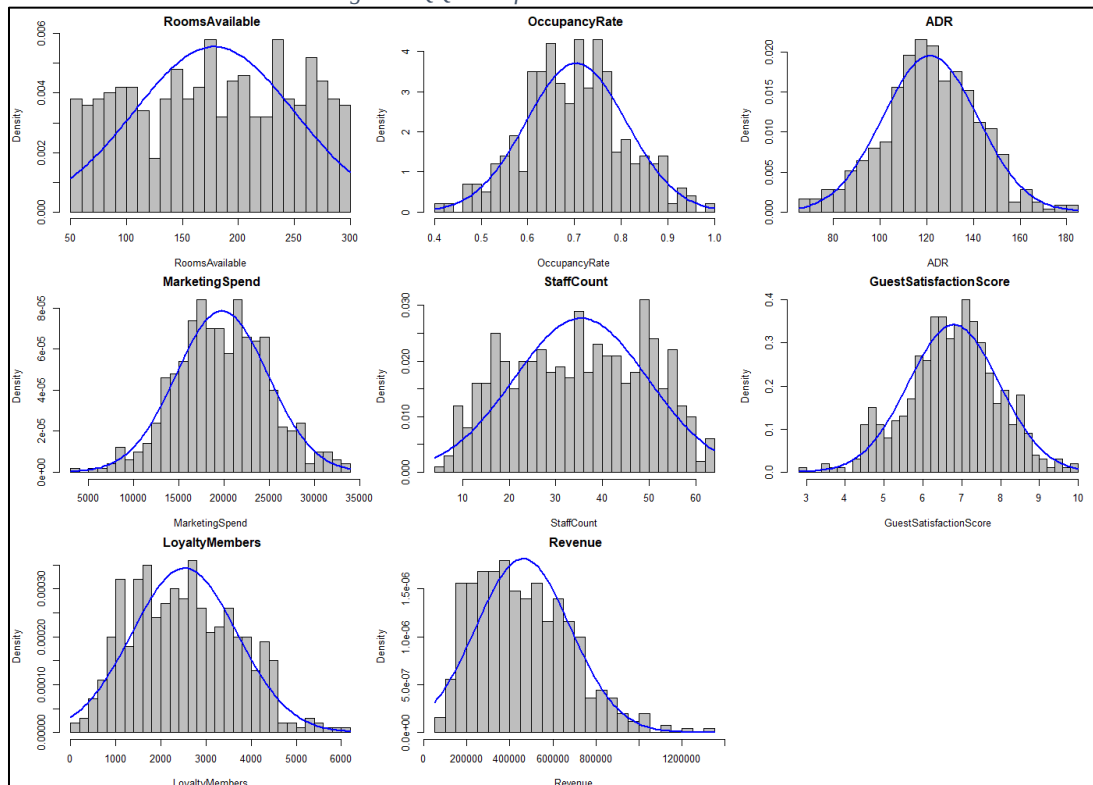


Figure 2 Histograms with Normality Curves for All Numerical Variables

QQ plots provide a visual check for linearity against expected normal quantiles, while histograms show the frequency distribution overlaid with the theoretical normal curve.

3.4. Interpretation of Normality Tests

The normality assessments reveal mixed results across variables:

- Variables Following Normal Distribution (fail to reject H_0 ; $p > 0.05$):
 - Occupancy Rate
 - ADR
 - Marketing Spend
 - Guest Satisfaction Score
- Variables Deviating from Normality (reject H_0 ; $p < 0.05$):
 - Rooms Available
 - Staff Count
 - Revenue
 - Loyalty Members

Observations:

- **Revenue** shows positive skew (0.618), which is typical in hospitality data where a few hotels generate high revenue.
- Deviations from normality in Rooms Available, Staff Count, Loyalty Members are modest and can be handled by transformation or non-parametric methods if needed.
- The combination of **Shapiro-Wilk**, **Anderson-Darling**, **Lilliefors tests**, and visual inspection, provides robust confirmation of normality or its deviation.

4. Hypothesis based correlation analysis

Correlation analysis measures the strength and direction of relationships between variables to identify factors associated with revenue. Both **Pearson (parametric)** and **Spearman (non-parametric)** tests were applied.

4.1. Hypotheses

For each correlation test, the hypotheses are formulated as:

- H_0 : No correlation between the predictor and Revenue ($\rho = 0$)
- H_1 : A correlation exists ($\rho \neq 0$, two-tailed)

A p-value < 0.05 indicates **rejection of H_0** , suggesting a statistically significant correlation.

4.2. Correlation with Revenue

Predictor Variable	Pearson r	Spearman ρ	p-value (Pearson)	Strength
Loyalty Members	0.923	0.934	<0.001	Strong
Rooms Available	0.857	0.884	<0.001	Strong
Staff Count	0.835	0.870	<0.001	Strong
ADR	0.364	0.329	<0.001	Moderate
Occupancy Rate	0.309	0.275	<0.001	Moderate
Guest Satisfaction	0.174	0.156	<0.001	Weak
Marketing Spend	0.005	-0.003	0.908	None

4.3. Graphical Assessment

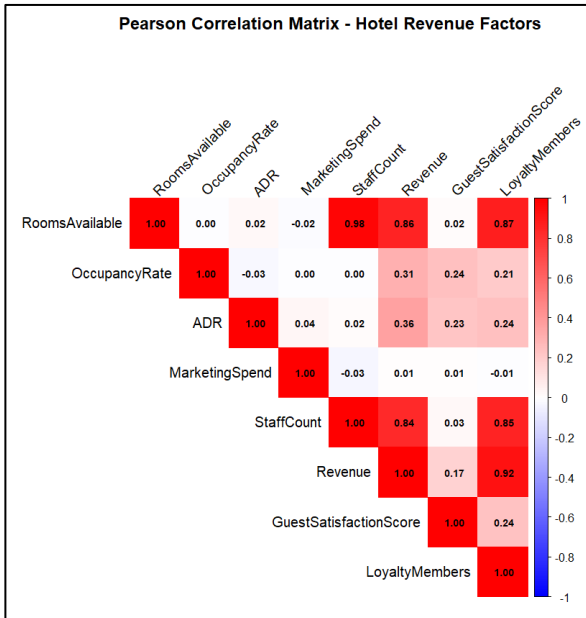


Figure 3 Correlation Heatmap for All Variables

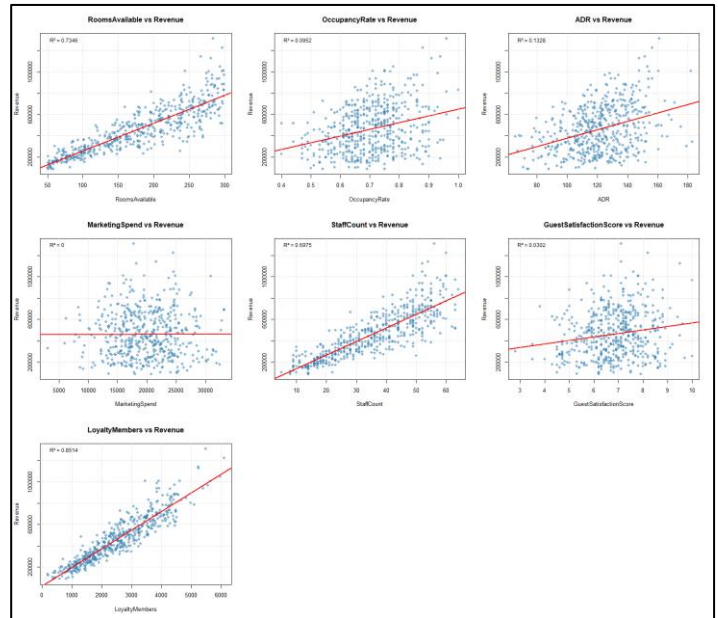


Figure 4 Scatter Plots of Revenue vs Numeric Predictors with Regression Lines

The heatmap provides a visual overview of pairwise correlations, highlighting strong and weak associations and helping identify potential multicollinearity among predictors.

The scatter plots show the relationship between each numeric predictor and Revenue. Blue points represent individual data observations, while the black lines show the linear regression trend. The plots visually confirm the direction and strength of relationships identified in the correlation tables and highlight potential non-linear patterns or outliers.

4.4. Key Findings

- **Loyalty Programs:** Loyalty Members has the strongest correlation with revenue ($r = 0.923$, $p < 0.001$), highlighting the high returns of investing in customer loyalty.
- **Operational Capacity:** Rooms Available ($r = 0.857$) and Staff Count ($r = 0.835$) are very strongly correlated with revenue, though they exhibit multicollinearity.

- **Pricing & Utilization:** ADR ($r = 0.364$) and Occupancy Rate ($r = 0.309$) show moderate correlations, indicating independent contributions to revenue.
- **Marketing Spend:** Shows negligible correlation ($r = 0.005$, $p = 0.908$), suggesting limited direct impact on revenue.
- **Guest Satisfaction:** Small but significant correlation ($r = 0.174$, $p < 0.001$), likely reflecting indirect effects on repeat business and referrals.

5. Regression analysis

A full-scale regression analysis was conducted using both simple and multiple linear regression models to evaluate predictors of hotel revenue.

1. Simple linear regression analysis

Simple linear regression was conducted to examine the individual effect of each predictor on hotel revenue at a 5% significance level and 95% confidence interval. The general model is:

$$\text{Revenue} = \beta_0 + \beta_1 X$$

Predictor	R ²	Adjusted R ²	RMSE	F-statistic	p-value	Significance
Loyalty Members	0.8514	0.8511	84,593.51	2852.80	< 0.001	Significant
Rooms Available	0.7346	0.7340	113,050.52	1378.19	< 0.001	Significant
Staff Count	0.6975	0.6969	120,686.84	1148.28	< 0.001	Significant
ADR	0.1328	0.1311	204,339.78	76.27	< 0.001	Significant
Occupancy Rate	0.0952	0.0934	208,727.42	52.38	< 0.001	Significant
Guest Satisfaction Score	0.0302	0.0283	216,088.37	15.52	< 0.001	Significant
Marketing Spend	0.0000	-0.0020	219,427.34	0.01	0.908	Not Significant

Table 4 Summary of Simple Linear Regression Models

❖ Analysis of Simple Regression Models

Best Single Predictor - Loyalty Members:

The model $\text{Revenue} \sim \text{Loyalty Members}$ achieved $R^2 = 0.851$, $\text{RMSE} = 84,763$, explaining 85.1% of revenue variance:

$$\text{Revenue} = 20,808.69 + 174.60 \times \text{Loyalty Members}$$

Each additional loyalty member increases revenue by **174.60 USD**, highlighting the importance of customer loyalty programs.

Diagnostic Plots:

Residual plots were generated to assess model assumptions:

- **Residuals vs Fitted:** Random scatter indicates linearity and constant variance.
- **Normal Q-Q:** Points closely follow the diagonal, supporting residual normality.
- **Scale-Location:** Residual spread is approximately constant.
- **Residuals vs Leverage:** No influential points exceed critical thresholds.

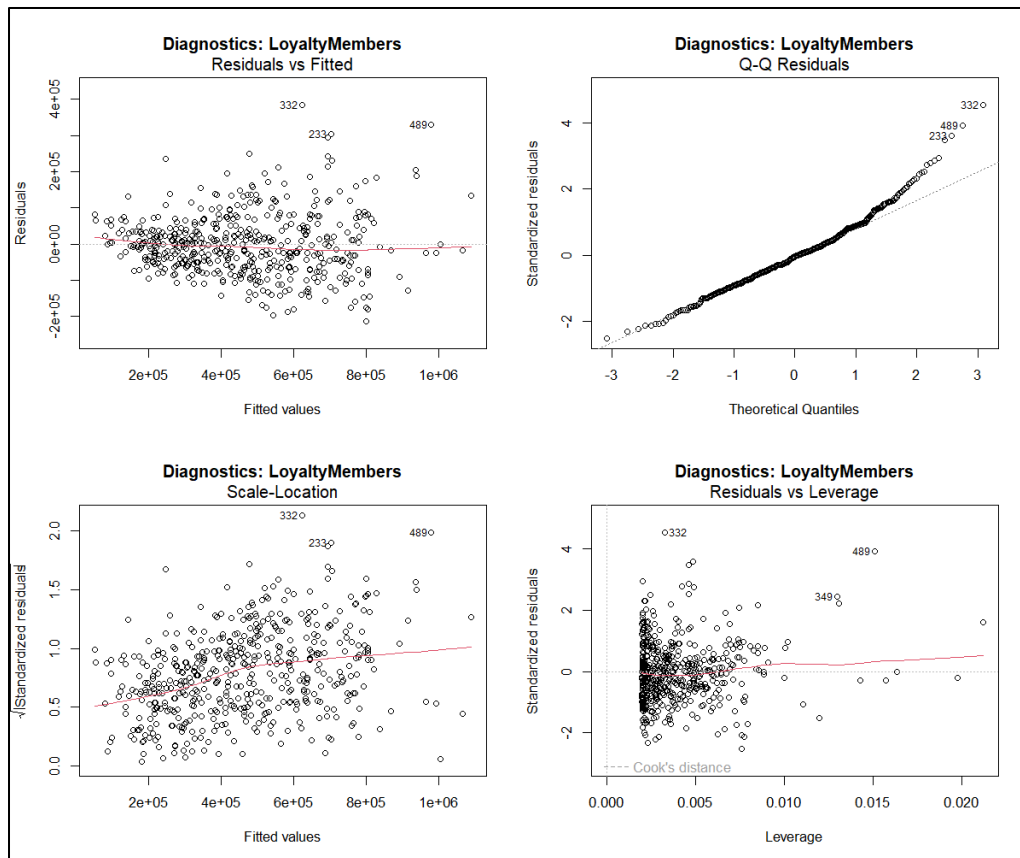


Figure 5 Diagnostic plots for the simple regression model with Loyalty Members as the predictor

Operational Capacity Variables:

Rooms Available ($R^2 = 0.735$) and Staff Count ($R^2 = 0.698$) show strong predictive power, but their high intercorrelation suggests overlapping constructs, requiring caution in multiple regression.

Moderate Predictors:

ADR ($R^2 = 0.133$) and Occupancy Rate ($R^2 = 0.095$) show moderate effects, with true contribution expected in multivariate models.

Non-Significant Predictor:

Marketing Spend ($R^2 \approx 0$, $p = 0.908$) shows no predictive validity, consistent with correlation analysis.

2. Multiple linear regression analysis

Multiple linear regression modeling was conducted through 3 iterations to identify the optimal predictive model for hotel revenue while addressing multicollinearity and exploring interaction effects.

❖ Model Development Strategy

Three models were developed and compared:

Model 1 – Full Model: Includes all seven numeric predictors to assess their simultaneous effects.

Model 2 – Refined Model: Retains only statistically significant predictors with acceptable multicollinearity ($VIF < 5$).

Model 3 – Enhanced Model: Incorporates interaction terms based on theoretical expectations of synergistic effects between room capacity, occupancy, and pricing.

❖ Multiple Regression Model Comparison

Model	Predictors	R ²	Adj R ²	RMSE	AIC	BIC
Model 1	7	0.9633	0.9628	42,386	10,596	10,634
Model 2	3	0.9576	0.9574	45,338	10,726	10,743
Model 3	5	0.9762	0.9759	34,091	10,443	10,468

Table 5 multiple regression model comparison

❖ Model 1 – Full Model (All Predictors)

$$\begin{aligned} \text{Revenue} = & -1,230,000 + 2,150 \times \text{RoomsAvailable} + 450,000 \times \text{OccupancyRate} \\ & + 3,800 \times \text{ADR} + 220 \times \text{MarketingSpend} - 459,050 \times \text{StaffCount} \\ & + 1,150 \times \text{GuestSatisfactionScore} + 8,100 \times \text{LoyaltyMembers} \end{aligned}$$

The full model explains 96.3% of revenue variance. However, high VIF values (RoomsAvailable = 28.79, StaffCount = 22.49, LoyaltyMembers = 8.16) indicate severe multicollinearity, making coefficient estimates unstable. MarketingSpend and StaffCount were non-significant ($p > 0.05$), reducing the model's interpretability.

❖ Model 2 – Refined Model (Core Predictors)

$$\text{Revenue} = -928,276 + 2,604 \times \text{RoomsAvailable} + 654,939 \times \text{OccupancyRate} + 3,827 \times \text{ADR}$$

After addressing multicollinearity, the model retains 3 key predictors as RoomsAvailable, OccupancyRate, and ADR. All predictors are highly significant ($p < 0.001$) with low VIF (< 1.01), indicating that hotel revenue is primarily driven by room capacity, occupancy, and pricing strategy.

❖ Model 3 – Enhanced Model with Interaction Terms (Best Model)

$$\begin{aligned} \text{Revenue} = & -117,181 - 45.92 \times \text{RoomsAvailable} - 478,959 \times \text{OccupancyRate} \\ & + 1,039 \times \text{ADR} + 3,734 \times (\text{RoomsAvailable} \times \text{OccupancyRate}) \\ & + 3,857 \times (\text{OccupancyRate} \times \text{ADR}) \end{aligned}$$

Key Coefficient Interpretations:

1. **RoomsAvailable $\times$ OccupancyRate ($\beta = 3,734$, $p < 0.001$):** The revenue impact of adding rooms increases substantially at higher occupancy rates, suggesting that expansion strategies should align with demand management.
2. **OccupancyRate $\times$ ADR ($\beta = 3,857$, $p < 0.001$):** Revenue is maximized when occupancy and ADR are high, supporting dynamic pricing strategies.
3. **Main Effects:** Negative coefficients for RoomsAvailable and OccupancyRate indicate that their positive contribution occurs primarily through interaction effects rather than independently.

❖ Conclusion

Model 3 is the most accurate and interpretable model, providing actionable insights for hotel management regarding capacity planning, occupancy management, and pricing strategies.

6. Discussion of the results and Implications for Sri Lanka's Tourism

The analysis highlights that hotel revenue in Sri Lanka is driven not by isolated factors but by the interaction of capacity, occupancy, pricing, and customer loyalty. These findings carry significant implications for both hotel management and national tourism policy.

6.1. Capacity-Occupancy Interaction

The strong positive interaction ($\beta = 3734.43$, $p < 0.001$) shows that revenue growth is amplified when both room capacity and occupancy increase. Hotels expanding capacity should target high-demand regions (Colombo, Galle, Mirissa) and align expansion with seasonal demand. Government infrastructure development near tourist hubs should be paired with strategic hotel growth to maximize revenue impact.

6.2. Pricing-Occupancy Dynamics

The occupancy-ADR interaction ($\beta = 3,856.73$, $p < 0.001$) validates dynamic pricing as a revenue lever. Hotels can increase ADR during peak periods while focusing on occupancy maintenance off-season. Mid-tier and boutique hotels, particularly, can benefit from training in revenue management and yield optimization.

6.3. Customer Loyalty Programs

Loyalty membership demonstrates the strongest revenue correlation ($r = 0.923$). Cross-property and airline-integrated programs, combined with Sri Lanka-specific benefits (temples, wildlife, tea plantations), can drive measurable revenue gains. Investment in digital CRM platforms ensures seamless booking and redemption, maximizing ROI.

6.4. Marketing Efficiency

The negligible correlation between marketing spend and revenue ($r = 0.005$) indicates inefficiencies. Strategic reallocation toward targeted digital campaigns, content marketing, and reputation management is recommended. Marketing attribution models can ensure budget effectiveness and revenue alignment.

6.5. Sector-Specific Recommendations

- **Cultural Tourism:** Leverage festivals and cultural experiences to justify premium pricing and improve occupancy.
- **Beach Resorts:** Align capacity expansion with peak-season demand and international marketing to attract high-ADR guests.
- **Eco-Tourism:** Boutique hotels should emphasize occupancy management and loyalty programs linked to conservation initiatives.
- **Business Hotels:** Stabilize revenue through corporate loyalty programs and conference facilities utilization.

6.6. Policy Implications

The findings support national initiatives including professional revenue management training, a cross-property hospitality loyalty framework, marketing budget optimization, and occupancy-based capacity planning. Tailored interventions for SMEs in rural areas are critical to ensure inclusive adoption of these strategies.

Key Takeaways:

1. Revenue is multiplicative therefore interaction effects dominate over individual factors.
2. Customer loyalty is the most impactful revenue driver.
3. Dynamic pricing and occupancy-aligned expansion offer high potential but require accurate forecasting.
4. Marketing inefficiencies highlight opportunities for data-driven reallocation.
5. Segment-specific strategies are essential for optimal revenue across cultural, beach, eco-tourism, and business hotels.

Screenshots of the statistical analysis workflow are included in [Appendix A](#).

The complete R project, along with all associated data and output files, is available at https://drive.google.com/drive/folders/1tYJN6KRR30i5HHQX7Hfzc0lshVadLR-N?usp=drive_link

Task B: Visualization of Tourist Arrivals to Sri Lanka – 2025

The objective of this task was to create a world map visualizing tourist arrivals to Sri Lanka from all countries during January–August 2025. The source data from Sri Lanka Tourism Development Board was provided as **All_Countries_Jan_August_2025.xlsx**.

1. Data Preparation

The original Excel dataset contained extensive and demographic and month wise information. To streamline the visualization, the dataset was filtered to keep only Country and Total columns. This CSV file, **SL_Tourists-2025.csv**, serves as the primary input for GIS processing.

The CSV was carefully compared against the attributes table of provided **world-administrative-boundaries.shp** file. Inconsistencies in country names (e.g., United States in the Excel file versus United States of America in the SHP) were **manually corrected** in the CSV to match the SHP file. This ensured that all available data would join correctly in QGIS.

Countries with no matching data in the CSV or missing from the SHP were later displayed in **grey** to indicate absence of arrival data.

2. Map Development

The map was generated in **QGIS** using a **join** between world-administrative-boundaries.shp and the cleaned SL_Tourists-2025.csv. **OpenStreetMap** served as the base map to provide geographic context. Total tourist arrivals were classified into **rule based categories** to reflect the relative contribution of each country.

To reduce clutter, only countries with more than **50,000 arrivals** were labeled directly on the map using the format “Country Name (Tourists arrival count)”.

Countries with missing data or no recorded arrivals were colored grey for clear differentiation. Standard cartographic elements including north arrow, numeric and graphic scale bars, map title, and legend were added to meet professional and academic standards.

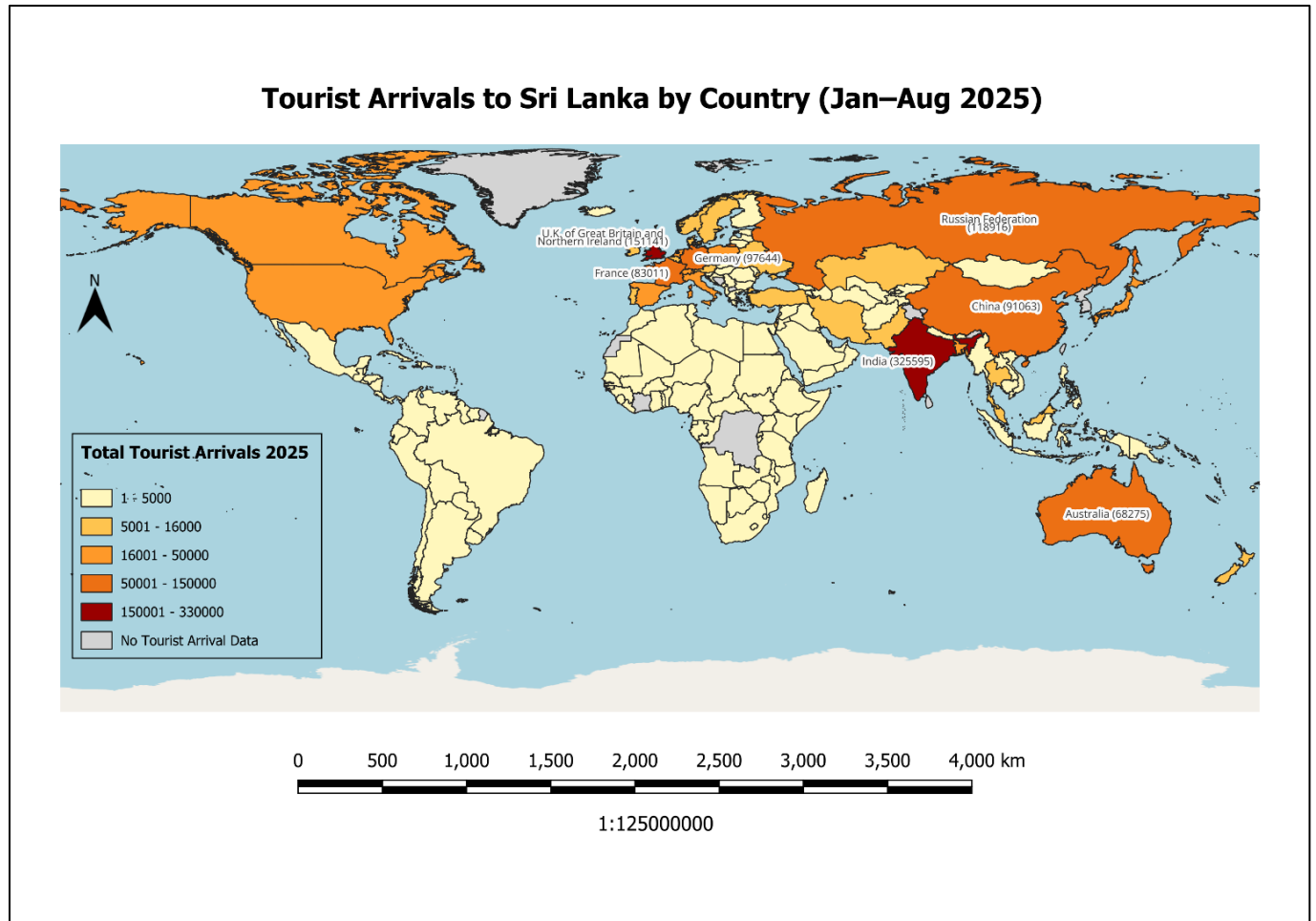


Figure 6 World Map of Tourist Arrivals to Sri Lanka, 2025

3. Map Analysis and Critical Discussion

The spatial distribution of tourist arrivals shows a strong concentration from a few key countries. **India** leads significantly with **325,595 visitors**, followed by the **United Kingdom (151,141)**, **Russian Federation (118,916)**, **Germany (97,644)**, **China (91,063)**, **France (83,011)**, and **Australia (68,275)**. These countries are labeled on the map for clarity. European countries like **France (83,011)**, **Italy (36,690)**, and **Netherlands (47,777)** also contributed substantial numbers, though only those exceeding 50,000 are labeled to maintain map clarity.

Medium-level arrivals come from countries such as **Bangladesh (40,485)**, **Canada (33,911)**, and **United States (44,605)**. Conversely, many countries, particularly in Africa, South America, and small island nations, had minimal arrivals, with some registering fewer than 10 tourists. These countries remain unlabeled but are visible in the rule based color scheme, allowing identification of both major and minor source countries. Grey-colored countries indicate regions with no data.

This visualization supports strategic tourism planning by highlighting high-value markets for infrastructure and marketing focus, while also identifying emerging markets for potential growth. The classification and selective labeling balance readability with detailed insight.

4. Map Quality

The final map adheres to high cartographic standards. OpenStreetMap provides a clear geographic base, and rule based symbology effectively conveys relative arrivals. Selective labeling minimizes clutter while emphasizing major source countries. All required map elements, including north arrow, scales, title, and legend, are present (Ingram, n.d.).

5. Conclusion

The developed map effectively visualizes tourist arrivals to Sri Lanka in 2025. Through careful data preparation, harmonization of country names, selective labeling, and professional cartographic techniques, the visualization clearly conveys spatial patterns and provides actionable insights for tourism planning and management.

Screenshots of the GIS workflow are included in [Appendix B](#).

The complete QGIS project, along with all associated data and output files, is available at [https://drive.google.com/drive/folders/1NCdmr11T2a-T9SVEcQGJymxjcNEpG5lS?usp=drive link](https://drive.google.com/drive/folders/1NCdmr11T2a-T9SVEcQGJymxjcNEpG5lS?usp=drive_link)

Task C: Development of a Digitized Informative Area Map of the Sri Lanka Tourism Development Authority (STDA) and Its Suburbs

1. Introduction

Geospatial technologies and Geographic Information Systems (GIS) play a critical role in spatial planning, infrastructure management, and institutional decision-making. Accurate georeferencing and digitization transform raw aerial imagery into structured spatial datasets that support analysis, visualization, and policy evaluation (Brown.edu, n.d.). In the context of tourism governance, spatial intelligence is particularly important for coordinating development activities, regulating service providers, and supporting sustainable urban–regional integration.

This study focuses on the development of a digitized informative area map of the Sri Lanka Tourism Development Authority (STDA) and its surrounding suburbs. The primary objective was to georeference the provided aerial image using Ground Control Points (GCPs) before initiating digitization, thereby ensuring positional accuracy under CRS EPSG:5234. The workflow was conducted using QGIS, with GCPs collected through Google Earth Pro and feature verification supported by Google Maps. Following successful georeferencing, thematic layers were digitized and structured with standardized attribute fields to produce a professional and analytically reliable cartographic output.

2. Methodology

2.1 Georeferencing

The aerial image was imported into the Georeferencer tool in QGIS. Seven Ground Control Points (GCPs) (A–G) were selected based on best-practice criteria:

- **Wide spatial distribution** to prevent distortion
- **Use of static man-made features**, primarily building corners
- Inclusion of a **jetty structure in the lake** for hydro-area stability
- High-contrast, visually distinct points for pixel-level precision

Points A and D anchored the western coastal edge, B the northeast, C the southeast, and E–G stabilized the central region (Brown.edu, n.d.) (QGIS project, 2025).



Figure 7 Selected GCPs

2.2 Digitization

Following successful georeferencing, four vector layers were digitized as

- **STDA Building** (Point)
- **Nearby Buildings** (Point)
- **Roads** (Line)
- **Hydro (Beira Lake)** (Polygon)

Each attribute table contained standardized fields as id, name, type, and size/area (where applicable).

The nearby buildings layer included hotels (Cinnamon Grand Hotel, Galle Face Hotel), offices (Ceylon Tourist Board, Ministry of Industry and Commerce), financial institutions (DFCC Bank – Head Office, Seylan Bank PLC), restaurants, and a school. Roads included Galle Road, Colombo Road, Sri Uttarananda Mawatha (main roads), Kirk Road (secondary), and Crescat Lane (residential). Hydro features represented three mapped segments of Beira Lake with measured areas.

2.3 Cartographic Layout

The final map layout incorporated map title, North arrow, graphic and numeric scale and the map legend.

Google Satellite map was included as the base map for contextual reference.

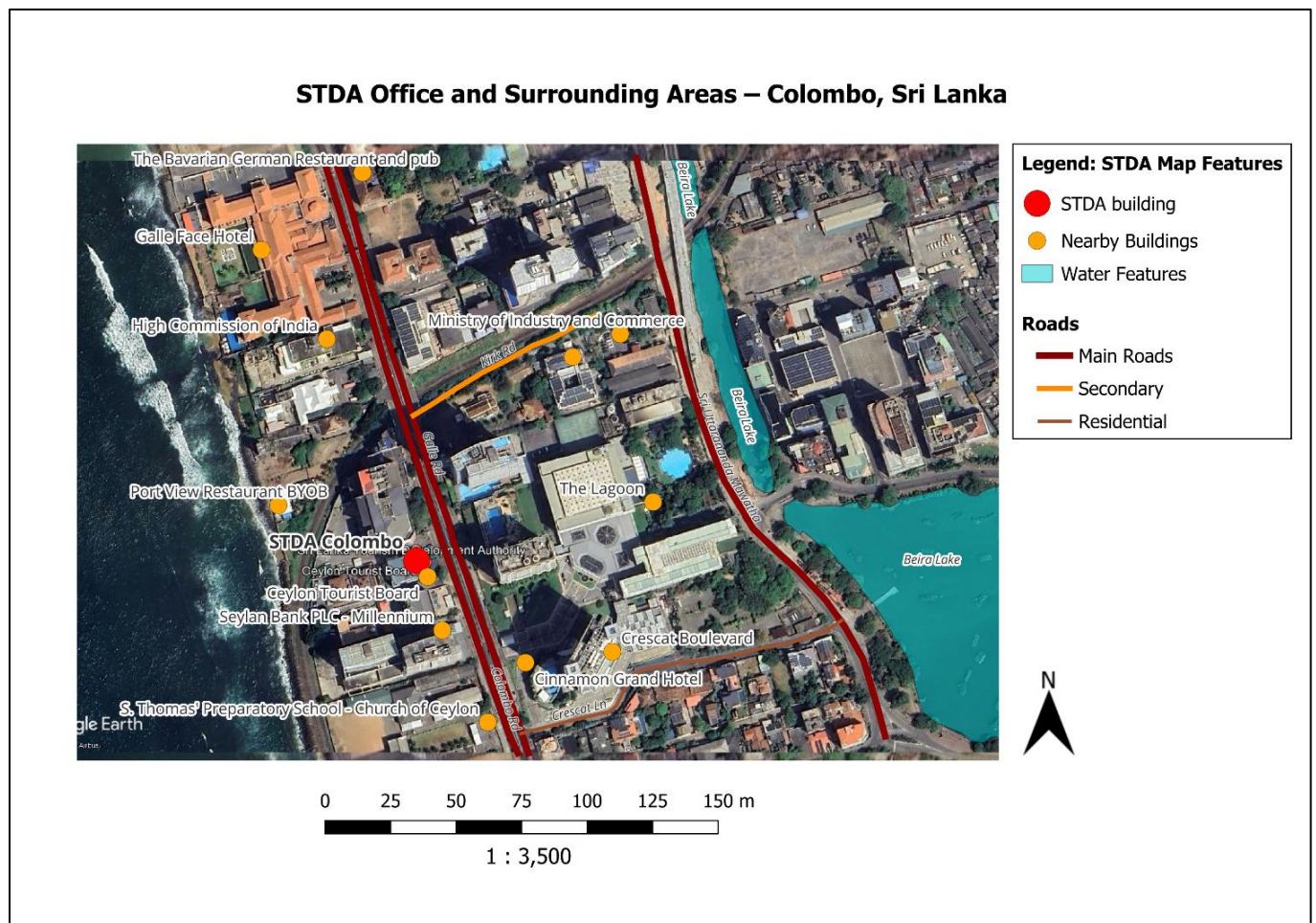


Figure 8 Digitized area map of STDA office and surrounding areas

3. Observations and analysis

The spatial configuration demonstrates that STDA is located within Colombo's administrative-tourism core. Proximity to government offices and financial institutions strengthens policy coordination, diplomatic engagement, and investment facilitation. The presence of major hotels, restaurants, and commercial complexes embeds STDA within an active tourism ecosystem, enabling direct stakeholder interaction and regulatory oversight.

Transport accessibility is a major advantage. The dominance of main roads (e.g., Galle Road and Colombo Road) ensures high connectivity, facilitating efficient administrative operations and stakeholder mobility. Additionally, proximity to Beira Lake reflects the integration of tourism planning with urban environmental management.

From a national perspective, STDA's centralized urban location enhances governance efficiency and data consolidation for islandwide tourism development. However, reliance on centralization may limit rapid responsiveness to regional tourism challenges. A GIS-integrated decentralized support structure could further strengthen national operational effectiveness.

4. Conclusion

The project successfully georeferenced the aerial image and produced a comprehensive digitized area map of STDA and its suburbs under CRS **EPSG:5234**. Proper GCP selection ensured spatial accuracy, and all required thematic layers with standardized attributes were developed. The final map meets professional cartographic standards and supports critical spatial evaluation. The analysis confirms that STDA's strategic geolocation significantly enhances its administrative coordination, stakeholder engagement, and national tourism governance capacity.

Screenshots of the GIS workflow are included in [Appendix C](#).

The complete QGIS project, along with all associated data and output files, is available at https://drive.google.com/drive/folders/1xz9ibgw57VXwc4CcooIP9VCBwckb_Fjj?usp=drive_link

Task D: Development of a Tourism Development Zones Map of Sri Lanka Using PostGIS and QGIS

1. Introduction

This project designed and implemented a spatial tourism mapping and analysis system to visualize Tourism Development Zones identified by the Sri Lanka Tourism Development Authority (SLTDA). The system integrates PostgreSQL with PostGIS for spatial data management and QGIS for visualization, spatial analysis, and professional cartographic output.

The primary objective was to produce a standards-compliant tourism development map and a structured geospatial database (SL_TD_Zones_2025) that supports evidence-based tourism planning. The system enables accurate geolocation, spatial querying, infrastructure assessment, environmental evaluation, and representation of tourism zones across Sri Lanka.

2. Data Collection and Preparation

A total of **45** tourism development locations were identified based on official SLTDA publications (Sri Lanka Tourism Development Authority, n.d.).

Geographic coordinates were captured using **Google Earth Pro** through KML extraction, ensuring positional accuracy via direct latitude and longitude recording. Each tourism location contains,

- Location name
- District and province
- Tourism regions (South Coast Resort Region, High Country Resort Region etc.)
- Tourism type (Coastal, Urban, Heritage, Eco, Wildlife, Hill, etc.)
- Geographic coordinates

The KML file were imported into the spatial database and served as the foundational geospatial dataset for analysis and mapping.

3. Geospatial Database Development

A PostgreSQL database named **SL_TD_Zones_2025** was created and extended with **PostGIS** to enable spatial data types, indexing, coordinate reference management, and advanced spatial functions.

Tourism zones were stored as point geometries in EPSG:4326 (WGS 84). Provided supporting spatial layers including roads, railway networks, forests, lagoons, cities, and administrative boundaries were also imported to the database as separate **tables** to support spatial analysis.

Spatial indexing was implemented to optimize query performance, particularly for proximity and intersection analyses.

4. Data Structuring and Attribute Processing

Semi-structured textual descriptions from KML were normalized using SQL string processing techniques. Geographic coordinates were dynamically extracted using PostGIS functions such as ST_X and ST_Y.

A structured **database view** (tourism_zones_view) was created to separate raw data storage from analytical representation. This abstraction improved modularity, ensured attribute consistency, and enhanced query efficiency.

5. Spatial Analysis

Spatial analysis was performed using **PostgreSQL with PostGIS spatial extensions** to quantitatively evaluate infrastructure accessibility, environmental sensitivity, and tourism-type suitability of selected tourism zones.

All analyses were conducted in geographic coordinate system (EPSG:4326). Distance calculations were performed on the `geography` type to ensure accurate meter-based measurements.

The combined results of these analyses are presented in **Table 6**.

ID	Zone	Distance to Nearest City (m)	Distance to Nearest Railway (m)	Forest Status	Distance to Nearest Lagoon (m)	Tourism Type
1	Negombo	131.80	18,945.93	Outside	550.19	Coastal
2	Colombo	291.57	0.00	Outside	10,733.35	Urban / Coastal
3	Mount Lavinia	437.81	3,685.64	Outside	3,221.83	Coastal
4	Wadduwa	21.01	2,947.62	Outside	4,920.48	Coastal
5	Kalutara	461.73	0.00	Outside	10,368.81	Coastal
6	Beruwala	180.96	9,380.19	Outside	4,863.44	Coastal
7	Bentota	131.16	16,011.39	Outside	889.09	Coastal
8	Dedduwa	318.43	18,512.11	Outside	489.86	Coastal
9	Madu Ganga	164.61	30,347.95	Outside	2,236.72	Eco / Coastal
10	Balapitiya	46.12	29,638.69	Outside	641.00	Coastal
11	Ahungalla	28.35	29,702.78	Outside	699.99	Coastal
12	Hikkaduwa	93.05	12,988.06	Outside	862.90	Coastal
13	Galle	88.07	0.00	Outside	10,840.91	Heritage / Coastal
14	Unawatuna	278.77	1,543.46	Outside	14,235.84	Coastal
15	Koggala	26.59	9,129.18	Outside	21,901.97	Coastal
16	Weligama	295.27	10,930.94	Outside	34,380.92	Coastal
17	Mirissa	97.64	8,245.20	Outside	37,234.64	Coastal
18	Matara	49.01	0.00	Outside	31,280.86	Urban / Coastal
19	Tangalle	943.13	12,040.12	Outside	17,608.03	Coastal
20	Hambantota	161.78	0.00	Outside	7,138.61	Coastal / Eco

21	Tissamaharama	858.53	8,584.86	Inside Forest	462.83	Eco / Heritage
22	Arugambay	179.00	49,595.37	Outside	4,607.02	Coastal / Surfing
23	Pasikudah	524.67	22,106.67	Outside	16,691.73	Coastal
24	Trincomalee	3,071.92	4,004.65	Outside	16,049.88	Coastal / Heritage
25	Nilaveli	12.60	10,467.87	Outside	342.10	Coastal
26	Kalpitiya	357.63	20,174.67	Outside	545.09	Coastal / Eco
27	Marawila	361.73	41,703.30	Outside	2,992.25	Coastal
28	Waikkal	231.17	25,227.82	Outside	9,635.37	Coastal
29	Nuwara Eliya	282.00	0.00	Outside	—	Hill / Climate
30	Bandarawela	1,282.26	16,687.24	Outside	—	Hill / Climate
31	Maskeliya	3,230.59	15,044.74	Outside	—	Hill / Climate
32	Polonnaruwa	9,577.35	14,991.41	Inside	—	Heritage / Cultural
33	Habarana	4,934.65	15,866.12	Outside	—	Heritage / Eco
34	Sigiriya	839.55	26,388.11	Outside	—	Heritage / Rock Fortress
35	Giritale	4,655.92	21,164.16	Outside	—	Heritage / Eco
36	Anuradhapura	1,646.26	524.95	Outside	—	Heritage / Cultural

37	Dambulla	3,008.82	22,898.85	Outside	—	Heritage / Cave Temples
38	Kandy	168.71	0.00	Outside	—	Heritage / Cultural
39	Matale	1,311.10	5,829.49	Outside	—	Heritage / Cultural
40	Victoria	1,687.21	11,614.41	Outside	—	Hill / Climate
41	Yala	12,530.56	46,993.51	Inside	15,132.17	Wildlife / Eco
42	Udawalawa	603.85	37,216.27	Outside	34,056.06	Wildlife / Eco
43	Wasgamuwa	1,339.11	28,865.92	Outside	—	Wildlife / Eco
44	Pinnawala	1,706.38	2,110.15	Outside	—	Wildlife / Cultural
45	Ratnapura	1,595.83	25,353.59	Outside	—	Urban / Gem Mining

Table 6 Summary of spacial analysis

5.1. Infrastructure Accessibility (Urban and Road Proximity)

Distance analysis measured each tourism zone's proximity to the nearest city and railway network.

Results indicate strong accessibility within the Western and Southern coastal corridor (Negombo to Matara). Many zones including Colombo, Kalutara, Galle, Matara, and Kandy have direct or near-direct railway connectivity (0 m distance), demonstrating high transport integration.

Coastal zones such as Wadduwa (21 m to city), Bentota (131 m), and Hikkaduwa (93 m) show exceptional urban proximity, confirming their infrastructure readiness and strong investment attractiveness.

In contrast, Eastern zones such as Arugambay (49 km to railway) and Pasikudah (22 km to railway) display weaker railway connectivity, indicating infrastructure gaps limiting tourism expansion.

Wildlife zones such as Yala (12.5 km to city; 47 km to railway) demonstrate intentional remoteness, aligning with conservation-based tourism models.

5.2. Railway Connectivity and Regional Balance

Railway accessibility is concentrated in Western and Southern regions, reinforcing their dominance in tourism flows. Inland heritage regions including Dambulla, Polonnaruwa, and Sigiriya rely heavily on road-based tourism circuits due to limited rail access.

This spatial imbalance highlights the **need for multimodal transport enhancement** to support equitable regional tourism development.

5.3. Environmental Sensitivity Analysis

Forest intersection analysis revealed three environmentally sensitive zones:

- Tissamaharama (Eco / Heritage)
- Polonnaruwa (Heritage / Cultural)
- Yala (Wildlife / Eco)

Yala's forest intersection combined with high remoteness confirms its ecological sensitivity and suitability for low-density, conservation-oriented tourism.

Tissamaharama's proximity to lagoons (462 m) and forest areas increases ecological vulnerability, requiring strict environmental regulation.

All other zones lie outside forest boundaries, reducing direct deforestation risk.

5.4. Lagoon Proximity and Coastal Suitability

Lagoon proximity strongly supports coastal and eco-tourism classifications. Zones such as Nilaveli (342 m), Kalpitiya (545 m), Dedduwa (489 m), and Negombo (550 m) demonstrate strong geographic alignment with their coastal tourism designation.

This confirms the spatial validity of SLTDA's tourism classification system.

6. Spatial Patterns and Development Contribution

The analysis reveals three major tourism spatial clusters:

1. Western–Southern Coastal Growth Corridor

Highly accessible, railway-integrated, and lagoon-adjacent. This corridor acts as Sri Lanka's primary tourism engine, contributing significantly to GDP and employment.

2. Cultural Triangle and Hill Country

Moderate accessibility with high cultural and climatic value. These zones require improved connectivity to maximize economic potential while protecting heritage assets.

3. Wildlife and Eco Regions

Spatially remote and environmentally sensitive. Their contribution lies in biodiversity conservation and high-value eco-tourism branding rather than mass tourism.

7. Map Design and Cartographic Output

The final thematic map was produced in QGIS using live database connectivity. Tourism zones were symbolized using categorized styling based on tourism type, improving interpretability.

Standard cartographic elements included map title, legend, North arrow and graphical and numeric scale bars.

Labels display location names, districts and latitude and longitude. The QGIS Print Layout tool was used to ensure professional cartographic standards.

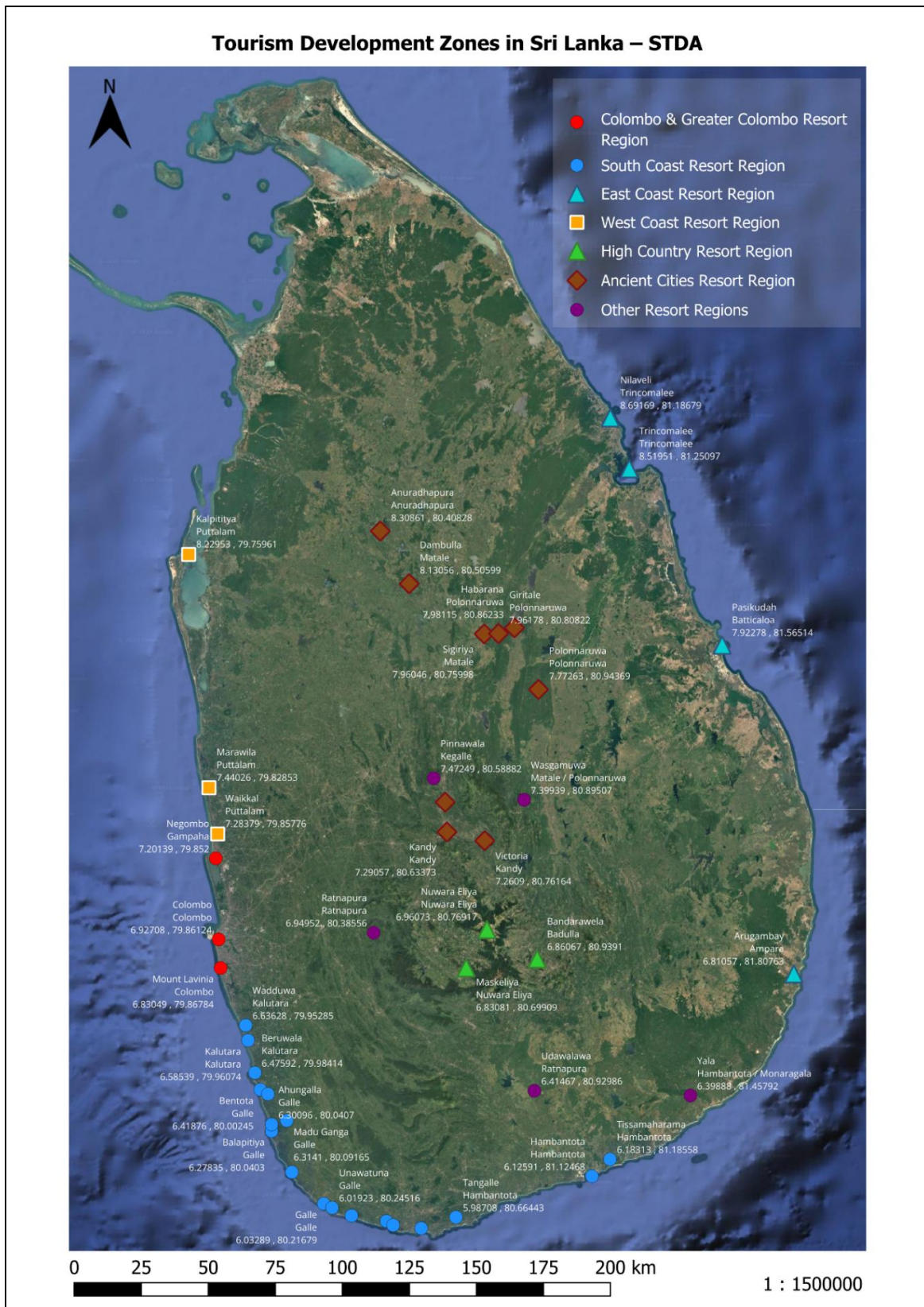


Figure 9 Map of tourism development zones in Sri Lanka

8. Discussion

Spatial analysis demonstrates that Sri Lanka's tourism development pattern is strongly influenced by infrastructure distribution and geographic advantages.

The Western and Southern coastal corridor benefits from superior accessibility and transport integration, reinforcing regional economic concentration. Meanwhile, Eastern coastal and wildlife zones face infrastructure limitations but possess high ecological and experiential value.

Forest-intersecting zones highlight the importance of balancing tourism growth with environmental conservation. The spatial distribution reflects strategic planning but also reveals regional disparities requiring targeted investment.

Thus, tourism development in Sri Lanka must adopt a spatially balanced and environmentally sensitive approach to ensure long-term sustainability.

9. Recommendations

1. Expand railway and road infrastructure toward Eastern Province tourism zones.
2. Strengthen multimodal transport links in Cultural Triangle regions.
3. Implement strict Environmental Impact Assessments for forest-adjacent zones.
4. Promote sustainable transport models (rail-based tourism, eco-mobility).
5. Maintain centralized geospatial monitoring systems using PostGIS.
6. Encourage community-based tourism to reduce regional inequality.

10. Conclusion

This project successfully integrates spatial database engineering and GIS cartography to support tourism planning in Sri Lanka.

The SL_TD_Zones_2025 spatial database and the final thematic map provide a scalable, data-driven framework for managing tourism development zones. Through spatial analysis, the system

reveals infrastructure strengths, regional imbalances, environmental sensitivities, and development opportunities.

The project demonstrates the practical application of,

- Spatial database technologies
- PostGIS analytical functions
- Coordinate reference system management
- GIS-based thematic visualization

Overall, the developed system supports informed, sustainable, and regionally balanced tourism policy planning while meeting professional cartographic standards.

Screenshots of the PostgreSQL and QGIS workflow are included in **Appendix D**.

The complete QGIS project, along with all associated data & output files and SQL scripts, is available at

https://drive.google.com/drive/folders/1a4H3yQ1JuJxPLC679eYFN6DDf9LkCsk?usp=drive_link

Task E: Identification of the Most Suitable Land for a Regional Tourism Development Center in Central Province, Sri Lanka

Note on Study Area

Although the assignment mentions the Central Province, the provided dataset corresponds to Gampaha District in the Western Province. This was later confirmed as a minor assignment error. Therefore, all analyses in this report are based on the provided dataset.

1. Introduction

This study identifies optimal land for the establishment of a Regional Tourism Development Center in Western Province, Sri Lanka, through the integration of spatial, environmental, and infrastructural criteria. Site suitability was evaluated based on the following considerations.

- **Proximity:** 2.4 km from Yagodamulla Maha Vidyalaya and 1.5 km from Kotugoda Roman Catholic School
- **Soil characteristics:** suitability for safe, stable, and cost-efficient construction
- **Agricultural preservation:** avoidance of areas cultivated with major export crops (tea, rubber, and coconut) to minimize disruption to productive land use
- **Environmental and land-use constraints:** exclusion of ecologically sensitive and built-up areas, including forests, water bodies, and existing urban infrastructure

The methodology integrates spatial analysis using QGIS with core GIS operations supplemented by verification through high-resolution satellite imagery. This multi-criteria approach supports a site selection that is accurate, feasible, and aligned with principles of sustainable regional development.

2. Data Sources

Data Layer	Type	Description
Buildings (CO_Building_Pg)	Polygon	Schools, residential, and other building structures

Administrative boundaries (AD_GN_Pg)	Arc & Polygon	Provincial and district boundaries
Land Use (LUSE)	Polygon	Land cover and soil classifications
Hydrography (HY_HYDRO_Pg)	Arc & Polygon	Rivers, streams, canals, tanks, reservoirs, ponds, lagoons
Hydrography (HY_HYDRO_Li)	Arc & Polygon	Boundaries of water features for buffer creation
Base Map	Raster	Google Satellite imagery for visual verification

Table 7 Summary of data sources used

All layers were projected to **EPSG:5234 (Kandawala/ Sri Lanka Grid)** to maintain spatial consistency.

3. Methodology

The workflow integrates **buffering, intersection, clipping, and difference** operations in QGIS, complemented by verification against a Google satellite basemap to ensure real-world relevance and feasibility.

The procedure followed a structured sequence of analytical steps, as described below.

3.1. Defining project coordinate reference system

The project coordinate reference system (CRS) was defined as EPSG:5234 (Kandawala / Sri Lanka Grid). All spatial layers were reprojected to this CRS to ensure positional consistency and enable accurate spatial analysis.

3.2. Buffering and proximity analysis

1. Yagodamulla Maha Vidyalaya and Kotugoda Roman Catholic School were selected from the building layer.
2. Proximity **buffers** were generated as 2.4 km for Yagodamulla Maha Vidyalaya and 1.5 km for Kotugoda Roman Catholic School.
3. The **Intersection** tool was applied to select spatial zones satisfying both proximity criteria simultaneously.

3.3. Administrative area representation

Using the administrative boundaries layer (AD_GN_Pg), the intersected area was verified to lie within Gampaha District of the Western Province, Sri Lanka. This confirmation ensured compliance with **administrative planning requirements**, therefore no additional intersection or difference operations were necessary to constrain the site to the designated province.

3.4. Exclusion of unsuitable land

Multiple categories of unsuitable land were identified and removed through spatial selection and overlay operations in QGIS. Land-use classes were determined using GFCODE classifications defined in the provided national geospatial data dictionary of Sri Lanka. Exclusion was implemented using the **Difference tool** to ensure that only technically feasible and regulation-compliant areas remained within the candidate suitability zone.

3.4.1. Traditional Export Crops

Areas under traditional export crop cultivation were excluded to prevent disruption to agricultural productivity and associated economic activities.

Procedure:

- Tea, rubber, and coconut cultivation areas were selected from the LUSE layer using the Query Builder.
- Selected areas were removed from the candidate suitability zone using the Difference tool.

These crops contribute substantially to regional livelihoods and export earnings. Their exclusion reduces the risk of economic displacement, land acquisition conflict, and long-term sustainability challenges.

3.4.2. Soil Suitability

Soil types unsuitable for structural foundations or safe construction were removed due to geotechnical instability, flood susceptibility, or disproportionate development cost.

GFCODE	Soil Type	Reason for Exclusion
MNGRA	Mangrove	Flood-prone, unstable organic soil
MRSHA	Marsh	Waterlogged conditions; high structural failure risk
SWMPA	Swamp	Low load-bearing capacity
ROCKA	Rock	Requires extensive excavation; high construction cost
QRRYA	Quarry	Uneven terrain; safety hazard
DSTSA	Distorted Surface	Irregular topography; unsuitable for foundations
SANDA	Sand Areas	Low cohesion; erosion and settlement risk
AQTCA	Aquatic Farms	Active aquaculture; ecological and economic conflict
HYDRA	Water Areas	Construction not feasible

Table 8 Summary of unsuitable soil types identified

The exclusion of geotechnically unstable soils enhances structural safety, reduces development costs, and minimizes long-term maintenance risk.

3.4.3. Other Unsuitable Land

Following land types were also identified as unsuitable for development.

Unsuitable land group	GFCODE	Land Type	Reason for Exclusion
Built-up and Social Areas	BLTPA	Built-up Area	Existing infrastructure; demolition and relocation required
	HOMSA	Homesteads / Gardens	Residential land; social disruption risk
	PARKA	Park	Public recreational space; protected land use
	PLGDA	Playground	Community infrastructure; cannot be repurposed
	CMTYA	Cemetery	Cultural, social and legal protection
Forests and Natural Vegetation	FRSDA	Dense Forest	High ecological value; deforestation impact
	FRSOA	Open Forest	Biodiversity and soil conservation function
	344 FRSPA	Forest Plantation	Economic forestry resource
	FRSUA	Forest – Unclassified	High environmental uncertainty
	SCRBA	Scrub Land	Erosion-prone if disturbed
	GRSLA	Grassland	Environmental buffering and flood regulation
Unclassified / Risk Areas	UNCLA	Unclassified	Unknown characteristics; planning risk

Table 9 Summary of other unsuitable land types identified

The removal of developed, protected, and ecologically sensitive land supports environmental sustainability, regulatory compliance, and social acceptability of the proposed development.

3.5. Hydrography Layer Exclusion

Hydrological features were excluded to ensure flood safety, environmental protection, infrastructure stability, and regulatory compliance. The HYDRO and HYDRO_Li layers represent

rivers, canals, tanks, reservoirs, lagoons, coastal features, and hydraulic infrastructure, including their spatial extents.

3.5.1. Hydrographic Buffers

Buffer distances for each feature were determined based on national land-use and development setback requirements as summarized below (Urban Development Authority, 2021).

GFCODE	Feature	Buffer Distance
STRML	Stream	4.5 m
CNNLL	Canal	6.5 m
CHNLL	Irrigation Channel	3 m
TANKL	Tank (minor)	15 m
RSVRL	Reservoir	45 m
LAGNL	Lagoon	30 m
OUTBL	Coastal Outline	25 m
BUNDL	Bund	5 m
DAML	Dam	50 m
Others	Minor Features	1 m

Table 10 Summary of buffer distance diagnosis

Using a **custom QGIS expression**, buffers were generated according to the GFCODE attribute in the vector layer, assigning the correct setback distance for each hydrographic feature type. Buffered areas were then removed from the candidate suitability zone using the **Difference** tool.

These buffer zones prevent development in areas susceptible to flooding or hydraulic failure, preserve aquatic ecosystems and natural drainage systems, ensure compliance with national environmental and land-use standards, and reduce long-term risks associated with infrastructure damage and flood mitigation costs.

The combined exclusion of unsuitable soils, land uses, and hydrographic buffers produced a **technically feasible, environmentally sustainable, and regulation-compliant suitability zone** suitable for regional tourism infrastructure development.

3.6. Building Analysis

High-resolution Google satellite imagery revealed several built-up structures absent from the provided building footprint layer. To address this incompleteness, **missing structures were digitized** as a supplementary building layer and integrated into the analysis, providing a conservative and realistic estimate of available land.

Then, the **Clip tool** was applied to get the buildings available within the final suitability zone. Then the area occupied by each building were computed using the QGIS `$area` expression and cross-validated with the **Measure tool** for accuracy.

This refinement ensured that land suitability assessments reflect current conditions and account for previously unrecorded built structures.

3.7. Final Suitability Zone

The final suitability zone was derived by removing unsuitable soil classes, incompatible land use categories, and predefined hydrographic buffer areas from the intersected buffer zone. The remaining area represents locations meeting all spatial suitability criteria.

The map was printed to include,

- Buffer center points and corresponding buffer areas
- The mapped out final suitability zone
- Buildings located within the zone
- Cartographic elements: North arrow, scale bar, numerical scale, legend, and map title

This visualization provides a clear representation of areas suitable for establishing a Regional Tourism Development Center in Western Province, Sri Lanka.

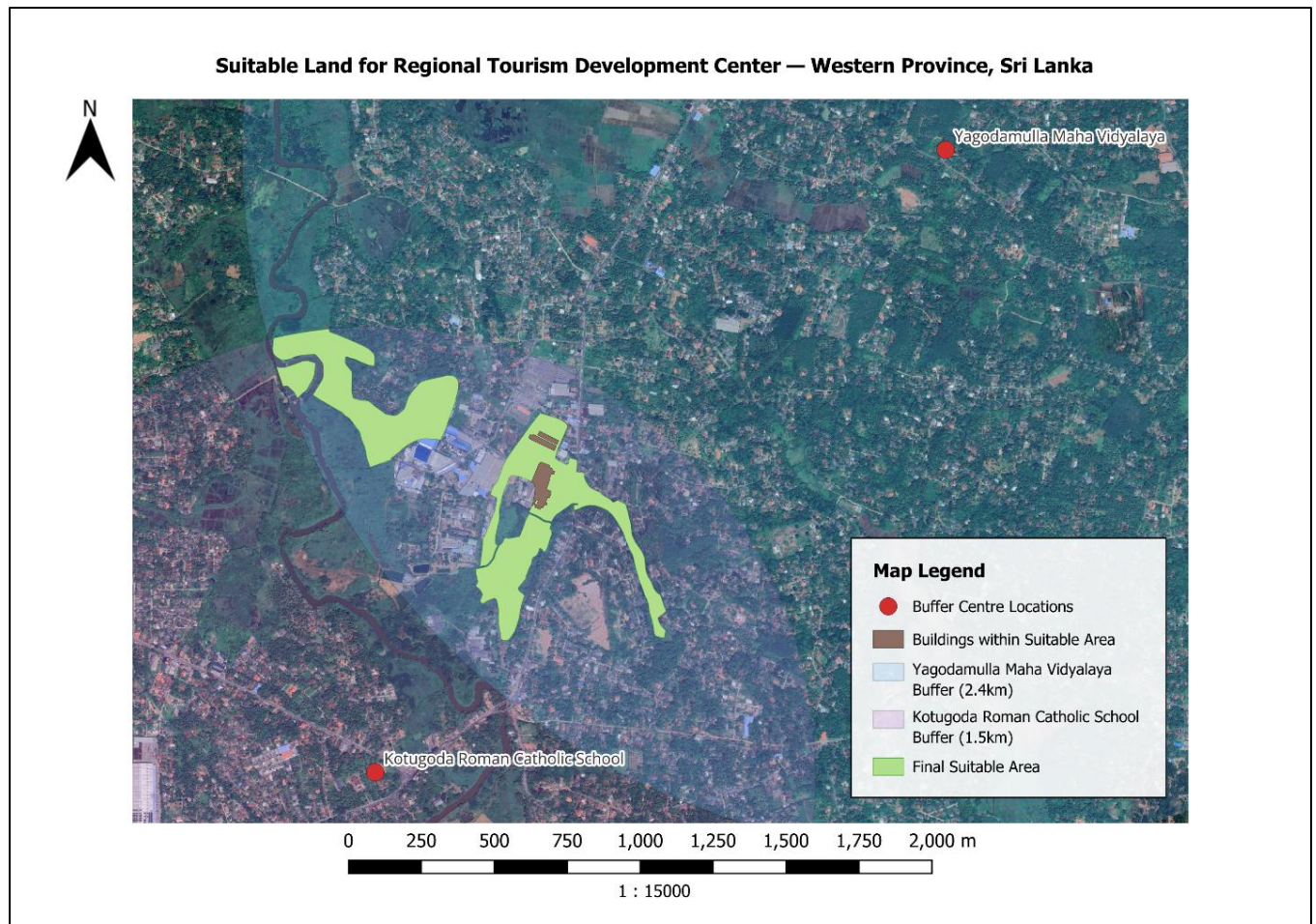


Figure 10 Map of buffer center locations, buffer areas, buildings available and final suitability zone identified

4. Results

Geospatial Identification

Centroid coordinates : 106029, 214489

Final suitability zone area : 251,101.55 m²

Building Analysis

Number of buildings within suitability zone : 3

Individual area footprints : 7357.67 m², 1097.28 m² and 1811.84 m²

Total land occupied by buildings : 10266.79 m²

Available Land

Total suitable land for construction: 241834.75 m² (24.18 hectares)

This indicates that sufficient space is available to accommodate the proposed Regional Tourism Development Center, including infrastructure, landscaping, and future expansion.

5. Analysis and observations

- **Proximity:** Buffers around nearby schools provide convenient access while ensuring the development integrates appropriately with the community.
- **Soil and land suitability:** Excluding unstable and unsuitable soils reduces structural risk and keeps construction costs manageable.
- **Water and environmental considerations:** Hydrography buffers help avoid flood-prone areas and maintain compliance with environmental regulations.
- **Infrastructure assessment:** Incorporating newly digitized buildings ensures that land-use estimates are realistic and account for existing structures.
- **Economic factors:** Avoiding areas with export crops or protected lands reduces potential compensation costs and preserves valuable resources.
- **Potential constraints:** Some unclassified areas may present unexpected limitations, and further on-site surveys, geotechnical studies, and evaluation of utility access are recommended before final development.

Overall, the site selection process combines environmental, social, and economic considerations to support a feasible and sustainable development.

6. Conclusion

The study identifies a safe, feasible, and legally suitable zone for establishing the Regional Tourism Development Center. By integrating soil, land use, hydrography, and building data, the analysis ensures that development can proceed sustainably, with minimal risk to the environment and existing land uses. The suitability zone provides adequate space for infrastructure, landscaping, and potential future expansion, offering a solid foundation for informed planning and decision-making.